Numerical Analysis: Exercises and Solutions

Computational Methods for Mathematical Problems
Mathematics 101

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January 18, 2026

Contents

1 Errors and Approximations

1.1 Error Calculations

Exercise 1.1. The exact value of $\sqrt{2}$ is 1.41421356.... If we approximate it as $\tilde{x} = 1.414$, calculate:

(a) The absolute error

(b) The relative error (as a percentage)

Solution

Let $x = 1.41421356$ and $\tilde{x} = 1.414$.

(a) Absolute error:

$$E_a = |x - \tilde{x}| = |1.41421356 - 1.414| = \boxed{0.00021356}$$

(b) Relative error:

$$E_r = \frac{|x - \tilde{x}|}{|x|} = \frac{0.00021356}{1.41421356} \approx 0.000151 = \boxed{0.0151\%}$$

Exercise 1.2. A measurement device reports a length of 15.7 cm with a relative error of 2%. Find the range of possible true values.

Solution

Relative error $= 2\% = 0.02$
Absolute error $= 0.02 \times 15.7 = 0.314$ cm
True value range: 15.7 ± 0.314
$\boxed{[15.386, 16.014] \text{ cm}}$

Exercise 1.3. Compute $f(x) = \sqrt{x+1} - \sqrt{x}$ for $x = 10000$ using:

(a) Direct calculation (using 6 significant figures)

(b) The rationalized formula $\frac{1}{\sqrt{x+1}+\sqrt{x}}$

Compare the results and explain which is more accurate.

Solution

(a) Direct calculation:

$$\begin{aligned} \sqrt{10001} &\approx 100.005 \\ \sqrt{10000} &= 100.000 \\ \sqrt{10001} - \sqrt{10000} &\approx 0.005 \end{aligned}$$

(Lost significant figures due to catastrophic cancellation)

(b) Rationalized formula:

$$\frac{1}{\sqrt{10001}+\sqrt{10000}} = \frac{1}{100.005+100} = \frac{1}{200.005} \approx \boxed{0.00499987}$$

The rationalized formula is more accurate because it avoids subtracting nearly equal numbers. The direct method loses precision due to catastrophic cancellation.

Exercise 1.4. How many iterations of the bisection method are needed to guarantee an error less than 10^{-6} starting with interval $[0, 1]$?

Solution

After n iterations, the error bound is $\frac{b-a}{2^{n+1}}$.
We need:

$$\frac{1-0}{2^{n+1}} < 10^{-6}$$

$$2^{n+1} > 10^6$$

$$(n+1)\log 2 > 6 \log 10$$

$$n+1 > \frac{6}{0.301} \approx 19.93$$

$$n \geq 19$$

$\boxed{n = 19 \text{ iterations}}$

2 Root Finding Methods

2.1 Bisection Method

Exercise 2.1. Use the bisection method to find a root of $f(x) = x^3 - x - 1$ in the interval $[1, 2]$. Perform 4 iterations.

Solution

First verify: $f(1) = 1 - 1 - 1 = -1 < 0$ and $f(2) = 8 - 2 - 1 = 5 > 0$. ✓

n	a	b	$c = \frac{a+b}{2}$	$f(c)$	New interval
1	1.0	2.0	1.5	1.875	$[1, 1.5]$
2	1.0	1.5	1.25	-0.297	$[1.25, 1.5]$
3	1.25	1.5	1.375	0.599	$[1.25, 1.375]$
4	1.25	1.375	1.3125	0.106	$[1.25, 1.3125]$

After 4 iterations: $\boxed{r \approx 1.3125}$ with error $\leq \frac{1}{2^5} = 0.03125$

Exercise 2.2. Show that $f(x) = e^x - 3x = 0$ has a root in $[0, 1]$ and find it using 3 iterations of bisection.

Solution

$f(0) = e^0 - 0 = 1 > 0$
$f(1) = e^1 - 3 = 2.718 - 3 = -0.282 < 0$
Sign change confirms root exists in $[0, 1]$.

n	a	b	c	$f(c)$	New interval
1	0	1	0.5	0.148	$[0.5, 1]$
2	0.5	1	0.75	-0.133	$[0.5, 0.75]$
3	0.5	0.75	0.625	-0.007	$[0.5, 0.625]$

$\boxed{r \approx 0.625}$

2.2 Newton-Raphson Method

Exercise 2.3. Use Newton-Raphson to find $\sqrt{3}$ by solving $f(x) = x^2 - 3 = 0$. Start with $x_0 = 2$ and perform 3 iterations.

Solution

$f(x) = x^2 - 3$, $f'(x) = 2x$
Newton-Raphson formula: $x_{n+1} = x_n - \frac{x_n^2-3}{2x_n} = \frac{x_n^2+3}{2x_n} = \frac{1}{2}\left(x_n + \frac{3}{x_n}\right)$

$$\begin{aligned}
x_0 &= 2 \\
x_1 &= \frac{1}{2}\left(2 + \frac{3}{2}\right) = \frac{1}{2}(3.5) = 1.75 \\
x_2 &= \frac{1}{2}\left(1.75 + \frac{3}{1.75}\right) = \frac{1}{2}(1.75 + 1.714) = 1.7321 \\
x_3 &= \frac{1}{2}\left(1.7321 + \frac{3}{1.7321}\right) = 1.7320508...
\end{aligned}$$

$\sqrt{3} = 1.7320508...$
After 3 iterations: $\boxed{x_3 = 1.7320508}$ (7 correct decimal places!)

Exercise 2.4. Apply Newton-Raphson to $f(x) = x^3 - 2x - 5 = 0$ with $x_0 = 2$. Perform 3 iterations.

Solution

$f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$
$x_{n+1} = x_n - \frac{x_n^3-2x_n-5}{3x_n^2-2}$

$$
\begin{aligned}
x_0 &= 2 \\
f(2) &= 8 - 4 - 5 = -1 \\
f'(2) &= 12 - 2 = 10 \\
x_1 &= 2 - \frac{-1}{10} = 2.1 \\
f(2.1) &= 9.261 - 4.2 - 5 = 0.061 \\
f'(2.1) &= 13.23 - 2 = 11.23 \\
x_2 &= 2.1 - \frac{0.061}{11.23} = 2.0946 \\
f(2.0946) &= 9.191 - 4.189 - 5 = 0.002 \\
f'(2.0946) &= 13.16 - 2 = 11.16 \\
x_3 &= 2.0946 - \frac{0.002}{11.16} = 2.09455
\end{aligned}
$$

$\boxed{x_3 \approx 2.0946}$

Exercise 2.5. Why might Newton-Raphson fail for $f(x) = x^{1/3}$ starting at $x_0 = 1$?

Solution

$f(x) = x^{1/3}$, $f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$

Newton-Raphson: $x_{n+1} = x_n - \frac{x_n^{1/3}}{\frac{1}{3}x_n^{-2/3}} = x_n - 3x_n = -2x_n$

Starting with $x_0 = 1$:

$$
\begin{aligned}
x_1 &= -2(1) = -2 \\
x_2 &= -2(-2) = 4 \\
x_3 &= -2(4) = -8 \\
&\vdots
\end{aligned}
$$

The method **diverges** because $|x_n| \to \infty$. This happens because the tangent line at any point (except $x = 0$) intersects the x-axis farther from the root than the starting point. The function has a vertical tangent at the root $x = 0$.

2.3 Secant Method

Exercise 2.6. Use the secant method to find a root of $f(x) = x^3 - x - 1$ with $x_0 = 1$ and $x_1 = 2$. Perform 3 iterations.

Solution

$x_{n+1} = x_n - f(x_n)\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$

$f(1) = -1$, $f(2) = 5$

$$
\begin{aligned}
x_2 &= 2 - 5 \cdot \frac{2-1}{5-(-1)} = 2 - \frac{5}{6} = 1.1667 \\
f(1.1667) &= 1.587 - 1.167 - 1 = -0.580 \\
x_3 &= 1.1667 - (-0.580) \cdot \frac{1.1667 - 2}{-0.580 - 5} = 1.1667 - \frac{0.483}{5.58} = 1.2532 \\
f(1.2532) &= -0.283 \\
x_4 &= 1.2532 - (-0.283) \cdot \frac{1.2532 - 1.1667}{-0.283 - (-0.580)} = 1.3313
\end{aligned}
$$

$\boxed{x_4 \approx 1.3313}$

2.4 Fixed-Point Iteration

Exercise 2.7. Rearrange $x^3 - x - 1 = 0$ as $x = g(x)$ in two ways:

(a) $x = x^3 - 1$

(b) $x = (x+1)^{1/3}$

Test convergence for each starting from $x_0 = 1.5$ by checking $|g'(x)|$ at the root (approximately $r = 1.3247$).

Solution

(a) $g(x) = x^3 - 1$, $g'(x) = 3x^2$
At $r \approx 1.3247$: $|g'(1.3247)| = 3(1.3247)^2 \approx 5.26 > 1$
Will diverge (fails convergence criterion)

(b) $g(x) = (x+1)^{1/3}$, $g'(x) = \frac{1}{3}(x+1)^{-2/3}$
At $r \approx 1.3247$: $|g'(1.3247)| = \frac{1}{3}(2.3247)^{-2/3} \approx 0.19 < 1$
Will converge ✓

Iteration with $g(x) = (x+1)^{1/3}$:

$$
\begin{aligned}
x_0 &= 1.5 \\
x_1 &= (2.5)^{1/3} = 1.357 \\
x_2 &= (2.357)^{1/3} = 1.331 \\
x_3 &= (2.331)^{1/3} = 1.326
\end{aligned}
$$

Converging to $\boxed{r \approx 1.3247}$

3 Interpolation

3.1 Lagrange Interpolation

Exercise 3.1. Find the Lagrange interpolating polynomial through $(0, 1)$, $(1, 2)$, $(2, 5)$.

Solution

Lagrange basis polynomials:

$$L_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}$$
$$L_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = \frac{x(x-2)}{-1} = -x(x-2)$$
$$L_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}$$

Interpolating polynomial:

$$\begin{aligned} P_2(x) &= 1 \cdot L_0(x) + 2 \cdot L_1(x) + 5 \cdot L_2(x) \\ &= \frac{(x-1)(x-2)}{2} - 2x(x-2) + \frac{5x(x-1)}{2} \\ &= \frac{x^2-3x+2}{2} - 2x^2 + 4x + \frac{5x^2-5x}{2} \\ &= \frac{x^2-3x+2-4x^2+8x+5x^2-5x}{2} \\ &= \frac{2x^2+2}{2} = \boxed{x^2+1} \end{aligned}$$

Verify: $P(0) = 1$, $P(1) = 2$, $P(2) = 5$ ✓

Exercise 3.2. Use Lagrange interpolation to estimate $f(2.5)$ given:

x	1	2	3	4
$f(x)$	0	1	8	27

(Note: The exact function is $f(x) = (x-1)^3$)

Solution

At $x = 2.5$:

$$L_0(2.5) = \frac{(2.5-2)(2.5-3)(2.5-4)}{(1-2)(1-3)(1-4)} = \frac{(0.5)(-0.5)(-1.5)}{(-1)(-2)(-3)} = \frac{0.375}{-6} = -0.0625$$
$$L_1(2.5) = \frac{(2.5-1)(2.5-3)(2.5-4)}{(2-1)(2-3)(2-4)} = \frac{(1.5)(-0.5)(-1.5)}{(1)(-1)(-2)} = \frac{1.125}{2} = 0.5625$$
$$L_2(2.5) = \frac{(2.5-1)(2.5-2)(2.5-4)}{(3-1)(3-2)(3-4)} = \frac{(1.5)(0.5)(-1.5)}{(2)(1)(-1)} = \frac{-1.125}{-2} = 0.5625$$
$$L_3(2.5) = \frac{(2.5-1)(2.5-2)(2.5-3)}{(4-1)(4-2)(4-3)} = \frac{(1.5)(0.5)(-0.5)}{(3)(2)(1)} = \frac{-0.375}{6} = -0.0625$$

$$\begin{aligned} P_3(2.5) &= 0(-0.0625) + 1(0.5625) + 8(0.5625) + 27(-0.0625) \\ &= 0 + 0.5625 + 4.5 - 1.6875 \\ &= \boxed{3.375} \end{aligned}$$

Exact: $(2.5-1)^3 = 1.5^3 = 3.375$ ✓

3.2 Newton's Divided Differences

Exercise 3.3. Construct the divided difference table for the data:

x	0	1	2	4
$f(x)$	1	1	2	5

Write the Newton interpolating polynomial.

Solution

Divided difference table:

x_i	$f[x_i]$	$f[\cdot,\cdot]$	$f[\cdot,\cdot,\cdot]$	$f[\cdot,\cdot,\cdot,\cdot]$
0	1			
		$\frac{1-1}{1-0} = 0$		
1	1		$\frac{1-0}{2-0} = \frac{1}{2}$	
		$\frac{2-1}{2-1} = 1$		$\frac{-\frac{1}{6}-\frac{1}{2}}{4-0} = -\frac{1}{6}$
2	2		$\frac{\frac{3}{2}-1}{4-1} = \frac{1}{6}$	
		$\frac{5-2}{4-2} = \frac{3}{2}$		
4	5			

Wait, let me recalculate: $f[x_2, x_3] = \frac{5-2}{4-2} = \frac{3}{2}$, $f[x_1, x_2, x_3] = \frac{\frac{3}{2}-1}{4-1} = \frac{1}{6}$
$f[x_0, x_1, x_2, x_3] = \frac{\frac{1}{6}-\frac{1}{2}}{4-0} = \frac{-\frac{1}{3}}{4} = -\frac{1}{12}$
Newton's polynomial:

$$
\begin{aligned}
P_3(x) &= f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\
&\quad + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2) \\
&= 1 + 0(x) + \frac{1}{2}x(x-1) - \frac{1}{12}x(x-1)(x-2)
\end{aligned}
$$

$$\boxed{P_3(x) = 1 + \frac{1}{2}x(x-1) - \frac{1}{12}x(x-1)(x-2)}$$

4 Numerical Differentiation and Integration

4.1 Numerical Differentiation

Exercise 4.1. Given $f(x) = \sin(x)$, approximate $f'(0.5)$ using $h = 0.1$:

(a) Forward difference

(b) Central difference

Compare with the exact value.

Solution

$f(0.4) = \sin(0.4) = 0.38942$, $f(0.5) = \sin(0.5) = 0.47943$, $f(0.6) = \sin(0.6) = 0.56464$

(a) Forward difference:

$$f'(0.5) \approx \frac{f(0.6) - f(0.5)}{0.1} = \frac{0.56464 - 0.47943}{0.1} = \boxed{0.8521}$$

(b) Central difference:

$$f'(0.5) \approx \frac{f(0.6) - f(0.4)}{0.2} = \frac{0.56464 - 0.38942}{0.2} = \boxed{0.8761}$$

Exact: $f'(x) = \cos(x)$, so $f'(0.5) = \cos(0.5) = 0.87758$
Central difference error: $|0.8761 - 0.8776| = 0.0015$
Forward difference error: $|0.8521 - 0.8776| = 0.0255$
Central difference is more accurate (as expected, since it's $O(h^2)$ vs $O(h)$).

Exercise 4.2. Approximate $f''(1)$ for $f(x) = e^x$ using the central difference formula with $h = 0.1$.

Solution

$f''(x) \approx \frac{f(x+h)-2f(x)+f(x-h)}{h^2}$
$f(0.9) = e^{0.9} = 2.4596$, $f(1) = e^1 = 2.7183$, $f(1.1) = e^{1.1} = 3.0042$

$$f''(1) \approx \frac{3.0042 - 2(2.7183) + 2.4596}{0.01} = \frac{0.0272}{0.01} = \boxed{2.72}$$

Exact: $f''(x) = e^x$, so $f''(1) = e = 2.7183$
Error: $|2.72 - 2.7183| = 0.002$ ✓

4.2 Numerical Integration

Exercise 4.3. Approximate $\int_0^1 e^{-x^2}\,dx$ using:

(a) Trapezoidal rule with $n = 4$

(b) Simpson's rule with $n = 4$

Solution

$h = \frac{1-0}{4} = 0.25$
Points: $x_0 = 0$, $x_1 = 0.25$, $x_2 = 0.5$, $x_3 = 0.75$, $x_4 = 1$
Function values:

$$\begin{aligned} f(0) &= e^0 = 1 \\ f(0.25) &= e^{-0.0625} = 0.9394 \\ f(0.5) &= e^{-0.25} = 0.7788 \\ f(0.75) &= e^{-0.5625} = 0.5698 \\ f(1) &= e^{-1} = 0.3679 \end{aligned}$$

(a) Trapezoidal rule:

$$\begin{aligned} I &\approx \frac{h}{2}[f_0 + 2(f_1 + f_2 + f_3) + f_4] \\ &= \frac{0.25}{2}[1 + 2(0.9394 + 0.7788 + 0.5698) + 0.3679] \\ &= 0.125[1 + 4.576 + 0.3679] \\ &= 0.125 \times 5.944 = \boxed{0.7430} \end{aligned}$$

(b) Simpson's rule:

$$\begin{aligned} I &\approx \frac{h}{3}[f_0 + 4(f_1 + f_3) + 2f_2 + f_4] \\ &= \frac{0.25}{3}[1 + 4(0.9394 + 0.5698) + 2(0.7788) + 0.3679] \\ &= \frac{0.25}{3}[1 + 6.0368 + 1.5576 + 0.3679] \\ &= \frac{0.25}{3} \times 8.9623 = \boxed{0.7469} \end{aligned}$$

(The "exact" value is approximately 0.7468, so Simpson's rule is more accurate.)

Exercise 4.4. How many subintervals are needed for the trapezoidal rule to approximate $\int_0^2 x^2\,dx$ with error less than 0.01?

Solution

For trapezoidal rule, error $\leq \frac{(b-a)^3}{12n^2}|f''(\xi)|$ for some $\xi \in [a, b]$.
$f(x) = x^2$, $f''(x) = 2$

$$\begin{aligned} \frac{(2-0)^3}{12n^2} \cdot 2 &< 0.01 \\ \frac{16}{12n^2} &< 0.01 \\ \frac{4}{3n^2} &< 0.01 \\ n^2 > \frac{4}{0.03} &= 133.3 \\ n &> 11.5 \end{aligned}$$

$\boxed{n = 12 \text{ subintervals}}$

5 Systems of Linear Equations

5.1 Gaussian Elimination

Exercise 5.1. Solve using Gaussian elimination with back substitution:

$$\begin{cases} 2x + y - z = 8 \\ -3x - y + 2z = -11 \\ -2x + y + 2z = -3 \end{cases}$$

Solution

Augmented matrix:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array}\right]$$

$R_2 \leftarrow R_2 + \frac{3}{2}R_1$, $R_3 \leftarrow R_3 + R_1$:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 2 & 1 & 5 \end{array}\right]$$

$R_3 \leftarrow R_3 - 4R_2$:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 0 & -1 & 1 \end{array}\right]$$

Back substitution:

$$\begin{aligned} -z = 1 &\Rightarrow z = -1 \\ \frac{1}{2}y + \frac{1}{2}(-1) = 1 &\Rightarrow y = 3 \\ 2x + 3 - (-1) = 8 &\Rightarrow x = 2 \end{aligned}$$

$\boxed{x = 2, \quad y = 3, \quad z = -1}$

Exercise 5.2. Why is partial pivoting important? Solve this system with and without pivoting:

$$\begin{cases} 0.0001x + y = 1 \\ x + y = 2 \end{cases}$$

using 3-digit arithmetic.

Solution

Without pivoting:
Multiplier: $m = \frac{1}{0.0001} = 10000$
$R_2 \leftarrow R_2 - 10000 \cdot R_1$:

$$y - 10000y = 2 - 10000 \Rightarrow -9999y = -9998$$

In 3-digit arithmetic: $y \approx 1.00$, then $x = 2 - 1 = 1$

But actual: $y = \frac{9998}{9999} \approx 0.9999$, $x \approx 1.0001$
With pivoting (swap rows first):

$$\begin{cases} x + y = 2 \\ 0.0001x + y = 1 \end{cases}$$

Multiplier: $m = 0.0001$
$R_2 \leftarrow R_2 - 0.0001 \cdot R_1$:

$$y - 0.0001y = 1 - 0.0002 \Rightarrow 0.9999y = 0.9998$$

$y = 0.9999 \approx 1.00$, $x = 2 - 1 = 1.00$
Pivoting gives more stable results because it avoids division by small numbers.

5.2 Iterative Methods

Exercise 5.3. Apply 3 iterations of the Jacobi method to:

$$\begin{cases} 4x - y = 15 \\ -x + 5y = 10 \end{cases}$$

starting with $x_0 = 0$, $y_0 = 0$.

Solution

Rearranged:

$$x = \frac{15 + y}{4}, \quad y = \frac{10 + x}{5}$$

k	$x^{(k)}$	$y^{(k)}$
0	0	0
1	$\frac{15+0}{4} = 3.75$	$\frac{10+0}{5} = 2$
2	$\frac{15+2}{4} = 4.25$	$\frac{10+3.75}{5} = 2.75$
3	$\frac{15+2.75}{4} = 4.4375$	$\frac{10+4.25}{5} = 2.85$

After 3 iterations: $\boxed{x \approx 4.44, \quad y \approx 2.85}$
(Exact solution: $x = 5$, $y = 3$)

Exercise 5.4. For the same system, perform 3 iterations of Gauss-Seidel starting from $(0, 0)$.

Solution

Gauss-Seidel uses updated values immediately:

$$\begin{aligned}
k=1: \quad x^{(1)} &= \frac{15+y^{(0)}}{4} = \frac{15+0}{4} = 3.75 \\
y^{(1)} &= \frac{10+x^{(1)}}{5} = \frac{10+3.75}{5} = 2.75 \\
k=2: \quad x^{(2)} &= \frac{15+2.75}{4} = 4.4375 \\
y^{(2)} &= \frac{10+4.4375}{5} = 2.8875 \\
k=3: \quad x^{(3)} &= \frac{15+2.8875}{4} = 4.4719 \\
y^{(3)} &= \frac{10+4.4719}{5} = 2.8944
\end{aligned}$$

After 3 iterations: $\boxed{x \approx 4.47, \quad y \approx 2.89}$
Gauss-Seidel converges faster than Jacobi (closer to exact solution $x = 5$, $y = 3$).

6 Numerical Solutions of ODEs

6.1 Euler's Method

Exercise 6.1. Use Euler's method with $h = 0.2$ to approximate $y(1)$ for:

$$y' = y, \quad y(0) = 1$$

Compare with the exact solution.

Solution

$f(t, y) = y$, $h = 0.2$, steps from $t = 0$ to $t = 1$ (5 steps)
Euler: $y_{n+1} = y_n + h \cdot y_n = y_n(1 + h) = 1.2y_n$

n	t_n	y_n (Euler)	$y(t_n)$ (Exact)
0	0.0	1.0000	1.0000
1	0.2	1.2000	1.2214
2	0.4	1.4400	1.4918
3	0.6	1.7280	1.8221
4	0.8	2.0736	2.2255
5	1.0	2.4883	2.7183

Euler approximation: $\boxed{y(1) \approx 2.4883}$
Exact: $y = e^t$, so $y(1) = e \approx 2.7183$
Error: $|2.4883 - 2.7183| = 0.23$ (about 8.5%)

Exercise 6.2. Apply Euler's method with $h = 0.1$ to solve $y' = -2ty$, $y(0) = 1$ for $0 \leq t \leq 0.5$.

Solution

$f(t, y) = -2ty$

n	t_n	y_n	$y_{n+1} = y_n + 0.1(-2t_n y_n)$
0	0.0	1.0000	$1 + 0.1(0) = 1.0000$
1	0.1	1.0000	$1 + 0.1(-0.2) = 0.9800$
2	0.2	0.9800	$0.98 + 0.1(-0.392) = 0.9408$
3	0.3	0.9408	$0.9408 + 0.1(-0.5645) = 0.8844$
4	0.4	0.8844	$0.8844 + 0.1(-0.7075) = 0.8136$
5	0.5	0.8136	–

$\boxed{y(0.5) \approx 0.8136}$

Exact: $y = e^{-t^2}$, $y(0.5) = e^{-0.25} = 0.7788$

6.2 Runge-Kutta Methods

Exercise 6.3. Use RK4 with $h = 0.2$ to approximate $y(0.2)$ for:

$$y' = y - t^2 + 1, \quad y(0) = 0.5$$

Solution

$f(t, y) = y - t^2 + 1$, $t_0 = 0$, $y_0 = 0.5$, $h = 0.2$

$$k_1 = hf(0, 0.5) = 0.2(0.5 - 0 + 1) = 0.2(1.5) = 0.3$$

$$k_2 = hf(0.1, 0.5 + 0.15) = 0.2(0.65 - 0.01 + 1) = 0.2(1.64) = 0.328$$

$$k_3 = hf(0.1, 0.5 + 0.164) = 0.2(0.664 - 0.01 + 1) = 0.2(1.654) = 0.3308$$

$$k_4 = hf(0.2, 0.5 + 0.3308) = 0.2(0.8308 - 0.04 + 1) = 0.2(1.7908) = 0.3582$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_1 = 0.5 + \frac{1}{6}(0.3 + 0.656 + 0.6616 + 0.3582) = 0.5 + \frac{1.9758}{6}$$

$$\boxed{y(0.2) \approx 0.8293}$$

Exercise 6.4. Compare Euler and RK4 for $y' = y$, $y(0) = 1$ at $t = 1$ using $h = 0.5$.

Solution

Exact: $y(1) = e \approx 2.7183$

Euler ($h = 0.5$, 2 steps):

$$y_1 = 1 + 0.5(1) = 1.5$$
$$y_2 = 1.5 + 0.5(1.5) = 2.25$$

Error: $|2.25 - 2.7183| = 0.468$
RK4 ($h = 0.5$, 2 steps):
Step 1 ($t_0 = 0$, $y_0 = 1$):

$$
\begin{aligned}
k_1 &= 0.5(1) = 0.5 \\
k_2 &= 0.5(1.25) = 0.625 \\
k_3 &= 0.5(1.3125) = 0.6563 \\
k_4 &= 0.5(1.6563) = 0.8281 \\
y_1 &= 1 + \frac{1}{6}(0.5 + 1.25 + 1.3125 + 0.8281) = 1.6487
\end{aligned}
$$

Step 2 ($t_1 = 0.5$, $y_1 = 1.6487$):

$$
\begin{aligned}
k_1 &= 0.5(1.6487) = 0.8244 \\
k_2 &= 0.5(2.0609) = 1.0305 \\
k_3 &= 0.5(2.1640) = 1.0820 \\
k_4 &= 0.5(2.7307) = 1.3653 \\
y_2 &= 1.6487 + \frac{1}{6}(0.8244 + 2.0609 + 2.164 + 1.3653) = 2.7181
\end{aligned}
$$

RK4 Error: $|2.7181 - 2.7183| = 0.0002$
Comparison:

Method	Approximation	Error
Euler	2.25	0.468
RK4	2.7181	0.0002

RK4 is dramatically more accurate with the same step size!